LAB 3

LAB 3 — IMPROVE A LAND-COVER CLASSIFIER

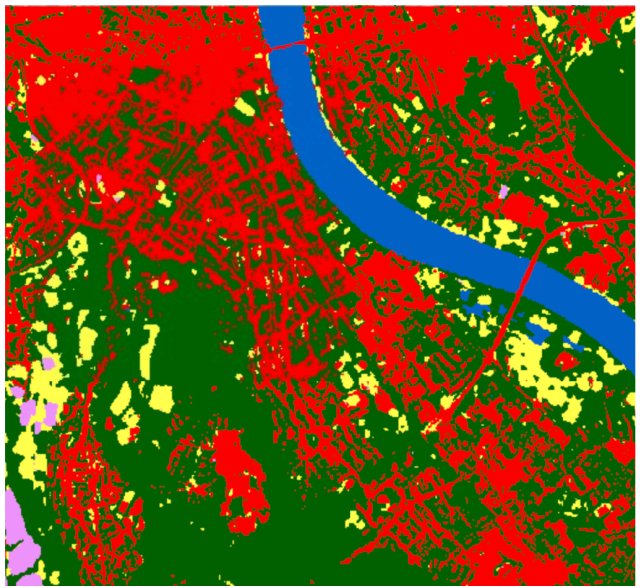
Start from the provided notebook and improve its accuracy.

PRESS → OR SPACE.

© Marc Rußwurm
Licensed under **Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0)**.
You may share and adapt these slides for teaching, research, and other non-commercial educational purposes with attribution.
Commercial training, paid workshops, consulting seminars, or incorporation into commercial course products require prior permission.

UNIVERSITÄT BONN

Task: Land-Cover Classification in Bonn



Build a **random forest** that predicts the land-cover class at each location — and improve its test accuracy **by any means necessary**:

1 · **More data** — add geodata from Lab 2

2 · **Decorrelated data** — PCA on the input variables

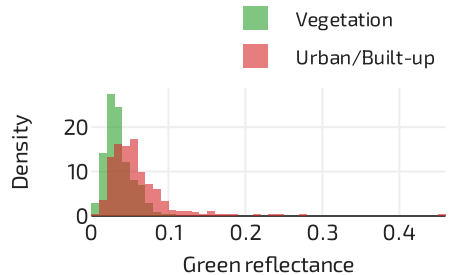
3 · **Domain-specific features** — e.g. NDVI

4 · **Better-tuned random forests** — hyperparameters

Dataset

Raw bands

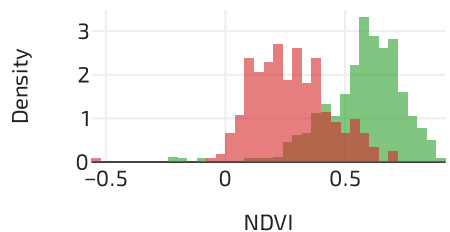
Pt	Red	Green	Blue	NIR	Land Cover
Pt1	0.08	0.11	0.09	0.42	**Vegetation**
Pt2	0.21	0.19	0.15	0.05	**Built-up**
⋮	⋮	⋮	⋮	⋮	⋮



Engineered feature

$$\mathrm{NDVI} = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + \mathrm{Red}}$$

Pt	NDVI	Land Cover
Pt1	0.68	**Vegetation**
Pt2	-0.62	**Built-up**
⋮	⋮	⋮



Your Task in Four Steps

1
Study the provided notebook

2
Improve the classifier by any of the four means

3
Run the evaluation and note the accuracy

4
Report your accuracy with your group number for the leaderboard

- **Notebook:** Lab 3 land-cover classification notebook — make a copy for your group
- **Lab 3 files:** Sentinel-2 image, land-cover raster and sample coordinates
- **Additional geodata:** Lab 2 data collection

Stay in your Lab 2 groups. Work with pandas; a CPU runtime is sufficient.

Step 1 · Study the Provided Notebook

Open the notebook, save a copy, and run it once from top to bottom with the Lab 3 files. Find out:

- **Locations:** where are the sample coordinates loaded, and which rows are `train` and `test`?
- **Features:** how are the Sentinel-2 bands sampled at these locations?
- **Labels:** how is the land-cover class sampled as the target?
- **DataFrame:** which columns form the features X and the target y?
- **Model:** how is the random forest fitted?
- **Evaluation:** where are the test accuracy and the confusion matrix computed?

Write down the **baseline accuracy** of the unchanged notebook. You will compare against it in Step 3.

Step 2 · Improve the Classifier

Try one or more of the four means — one change at a time, so you know what helped.

1 · More data
sample sources from the Lab 2 data collection at the same coordinates; add them as new columns, matched by `sample_id`

2 · Decorrelated data
transform the input columns with `sklearn.decomposition.PCA`; fit it on the `train` rows only

3 · Domain-specific features
compute indices such as NDVI from the bands; S2 bands: **B2, B3, B4, B8** (blue, green, red, NIR)

4 · Better-tuned random forests
e.g. `n_estimators`, `max_depth`, `max_features`; tune with cross-validation on the `train` rows

Checks: transform coordinates to each source's CRS (S2: EPSG:32632, land cover: EPSG:4326) · check missing values · never use the land-cover label or the `test` rows to build features or tune the model.

Step 3 · Run the Evaluation

Re-run the notebook's evaluation with your improved model.

- keep the supplied `train` / `test` split unchanged
- if your new data has gaps, evaluate the baseline and your model on the **same rows**

Model	Train rows	Test rows	Test accuracy
Baseline (Sentinel-2 only)	…	…	…
Your improved model (what you changed: …)	…	…	…

Look at the confusion matrix: which classes improve, and which errors remain?